

Summarizing Research Papers with BERT

Abstract

We present a lightweight framework that summarizes research papers using only their introduction and conclusion. Sentences are embedded with a pre-trained BERT encoder and the most representative ones are selected by clustering. Experiments on computer science papers show that summaries are more than sixty percent shorter while keeping the research problem, approach and findings.

1 Introduction

The volume of scientific literature is growing at an unprecedented rate, with millions of new research papers published every year. University repositories now contain lengthy documents that make it difficult for students and researchers to quickly understand the main contribution of a paper. Reading full papers to decide whether they are relevant is time consuming and often impractical. Automatic text summarization has therefore become an important tool for helping readers navigate this growing body of knowledge.

Traditional summarization techniques rely on hand-crafted features such as word frequency and sentence position, which fail to capture the deeper semantic meaning of scientific text. Recent advances in deep learning, and in particular transformer-based language models such as BERT, have shown a remarkable ability to represent the meaning of sentences in context. However, most existing summarization systems are trained on news articles and do not transfer well to the specialised vocabulary and structure of academic writing.

1.1 Contributions

In this paper, we propose a summarization approach that uses a pre-trained BERT model to encode the sentences of a paper and selects the most representative ones using clustering. We focus on the introduction and the conclusion because these sections state the research problem, the motivation, and the final outcomes of the work. Our goal is to produce short summaries that preserve the key research information without requiring any task-specific training data.

2 Related Work

[illegible]

3 Method

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4 Experiments

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5 Conclusion

In this work, we presented a BERT-based framework for summarizing research papers using only their introduction and conclusion. Our method embeds every sentence with a pre-trained BERT encoder and applies K-Means clustering to select the sentences closest to each cluster centre. Experiments on a collection of computer science papers show that the generated summaries reduce the text length by more than sixty percent while retaining the main research problem, approach, and findings. The results demonstrate that contextual embeddings capture the semantic structure of academic text far better than frequency-based methods. We also observed that combining information from both the introduction and the conclusion yields more complete summaries than using either section alone. Nevertheless, the extractive nature of the method means that some summaries can contain sentences that lack context when read in isolation. In future work, we plan to fine-tune the model on scientific corpora and to explore abstractive generation to produce more fluent summaries. Overall, we conclude that pre-trained language models provide a practical and effective solution for helping readers quickly grasp the contribution of a paper.

References

[1] Devlin et al. BERT. 2019.